



# Subtyping vs inheritance

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| ▼ Type        |  Lecture                                                                                                     |
| 📅 Date        | @January 8, 2022                                                                                                                                                                                |
| ☰ Lecture #   | 5                                                                                                                                                                                               |
| 🔗 Lecture URL | <a href="https://youtu.be/CYY7IT-YHVA">https://youtu.be/CYY7IT-YHVA</a>                                                                                                                         |
| 🔗 Notion URL  | <a href="https://21f1003586.notion.site/Subtyping-vs-inheritance-74043d6113e04d2faeab33d3706f824a">https://21f1003586.notion.site/Subtyping-vs-inheritance-74043d6113e04d2faeab33d3706f824a</a> |
| # Week #      | 3                                                                                                                                                                                               |

## Subclasses, subtyping and inheritance

- Class hierarchy provides both subtyping and inheritance
- Subtyping
  - Capabilities of the subtype are a superset of the main type
  - If `B` is a subtype of `A`, wherever we require an object of type `A`, we can use an object of type `B`
  - `Employee e = new Manager(...);` is legal
- Inheritance
  - Subtype can re-use code of the main type
  - `B` inherits from `A` if some functions for `B` are written in terms of functions of `A`
  - `Manager.bonus()` uses `Employee.bonus()`

## Subtyping vs inheritance

- Recall the following example
  - `queue`, with methods `insert-rear`, `delete-front`
  - `stack`, with methods `insert-front`, `delete-front`
  - `deque`, with methods `insert-front`, `delete-front`, `insert-rear`, `delete-rear`
- Subtyping
  - `deque` has more functionality than `queue` or `stack`
  - `deque` is a subtype of both these types
- Inheritance

- Can suppress two functions in a `deque` and use it as a `queue` or `stack`
- Both `queue` and `stack` inherit from `deque`